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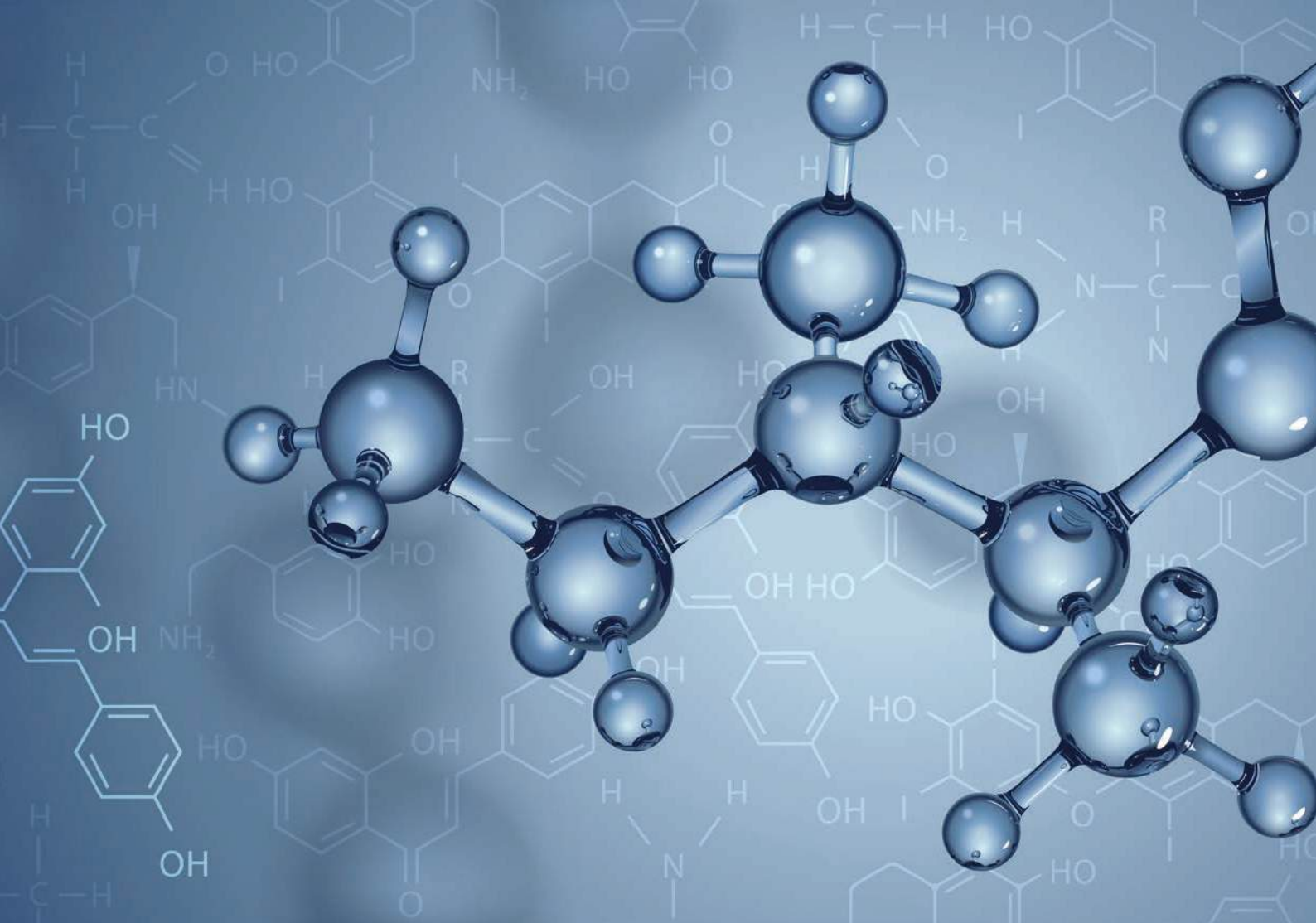
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Minimum Energy Structures: Organic Molecules as The Means to Solar Sustainability

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Sustainability has become a hot topic nowadays. From energy to even the smallest product, all consumers now want to know: “Is it sustainable?” With energy being the main thing that modern society needs to produce more of both cleanly and cheaply to reach the entire globe in the quest to make away with fossil fuel

dependency, organic molecules hold the solution to both these needs. As particles that can be determined best through experimentation, modern AI that can quickly test combinations is indeed a time saver, but it is most importantly the computational methods that “play a key role in the design of therapeutically important molecules

for modern drug development” [4], but are also very much applicable to making better molecules for a variety of other purposes as well. With AI as able to now power the computational power needed for initial quantum computers that can handle and thus make feasible a model of automated chemistry for helpful substances, solar cells as the next breakthrough in organic production will not only solve the upcoming issue of energy resources, but also avoid the current dangers that come from the substances needed to create artificial solar panels in the meantime.

Computational Drug Therapy: Synthesizing Wanted Molecules (?)

Existing in two broad categories that are either structure or ligand-based, the first “require knowledge of the structure of both the target and ligand” for calculation of available binding energy, while the second type “use only information about the ligand to predict the biological response depending on historical data about known active and inactive ligands” [4], both processes can assist in the making of energy evaluation for molecular systems, but still use either force fields or empirical scoring functions, which becomes almost unthinkable if millions of possible ligands need to be examined. Done through a “zoo” of force fields as developed over the last quarter century for applications in various domains, the functional results of these systems are a very difficult to diagnose when they fail, as: “(1) force-fields are fast, but can perform poorly outside of their fitting set, and

(2) quantum mechanical methods can be extremely accurate, but at a prohibitive computational cost” [4].

Quantum Mechanics: The Usable Properties

Lots of advances have become possible via atomistic potentials (and other properties) using machine learning (ML) methods, as “[t]he low numerical complexity and high accuracy of machine learning algorithms make them very attractive... [d]ue to their remarkable ability to find complex relationships among data, in most cases these “machine learned” models outperform more physically constrained approximations like force fields and semiempirical QM” [4]. Remembering that such ML methods are completely reliant on the data provided to them, Neural networks are another option that carry more options, for example, being highly efficient and effective at modeling reference training data in a flexible manner, but also have the drawback of needing a vast amount of reference data to prepare the model beforehand. Most true for general models that can provide accuracy for more than a limited few changes to the design variables, the authors give the example of their work with ANI-1 potential.

ANI1

The first example of a universal NN molecular potential, training via a large data set of 22 million small molecule conformations resulted in “a potential capable of predicting energies on much larger systems and on systems

wildly different from those in the training set” [4], a sure sign of future use in any framework that depends on prediction as a saver of both time and resources. With applicability for up to 70 atoms, including well-known drug molecules, the usability is even higher than might initially be hoped for, with a number as low as “0.57 kcal/mol when considering molecular conformations that are relevant at room temperature” when looking at RMSE versus DFT relative energies [4]. With the concept also workable in the context of piecing larger projects into smaller ones for ease and group work, as “training an ML model with small fragments of organic molecules to predict on larger systems has been verified independently” [4] via. Gastegger et al. who generated DFT data instantly for broken up larger organics, data sets as either large or fragmented bring possibility to the Feedback loop needed for synthesis of new molecules.

“Smart” Sampling - The Feedback Loop for ANI

Used in the field of molecular synthesis to check for results, newer methods work around specific approximations as predefined by data generated before the fitting process. ML methods combined with active learning (AL) as a natural analogy to the concept of (the artificial) feedback loop means that science now allows us to fit methods to nature, rather than try to adjust how we work contrary to the best way to do so. Recent improvements to ANI-1 came about based off an en-

semble of models (known as query by committee) to “drive diverse sampling methods and automatically select new molecules and conformations to add to the training set” that led to the ANI-1x as having “a very high-level of potential energy and force accuracy on large drug-like molecules and even protein sized systems from the COMP6 benchmark” [4].

As more accurate than the best small molecule force fields but also cheaper by magnitudes than competing advanced QM methods, ANI-1ccx potential as an attractive alternative to density functional theory approaches and standard force fields for conformational searches, as usable with transfer learning-based of a model that surpasses DFT on test cases, including those of failed reaction thermochemistry capture, quantum mechanics as the other possibility must not be completely discounted, as this approach “underpins all of modern chemistry” [3].

Quantum Computing and Mechanics: The Building Blocks as Energy Balancers

With the hypothetical belief that the method is costly but can predict any chemical compound for behavior, Dirac (1929) noted that “The exact application of these laws leads to equations much too complicated to be soluble,” as “the complexity of the wave function of a quantum system grows exponentially with the number of particles” [3]. Out of range of classic systems, quantum computing remains the modern solution to the problem, with several effi-

cient quantum algorithms currently proposed as possible solutions.

With run-time and physical resources needed for such algorithms scaling “polynomially with both the size of the system simulated and the accuracy required” [3], hardware capabilities might be for now solved by use of the above example of systems such as ANI1, which are not quantum, as a bridge to further developments in computing technology. Nevertheless, knowing what will be needed later is a necessity to any solid idea, as so far “no group has yet demonstrated a single fully error-corrected qubit” as is required, quantum algorithms as a solution to finding low-lying energy levels of chemical systems is the scope here. With the need arising from our effort to envision an organic solar cell that thus uses the least amount of activation energy, increasing its cellular efficiency to sunlight capture, the “electronic structure problem” is also valid as highly applicable to other types of problems in quantum computing, from finding transition states to comprehension of the vibrational structure of molecules [3].

For Solar Cells:

A Basis of Error Reduction

As “physical qubits accumulate errors during computation due to their interaction with the environment and our imperfect control”, while suppressed errors are possible but eat up much qubits used for computation, seen by Sawaya et al. (2016) when they “...used numerical simulations to show the impact that noise has on phase-estima-

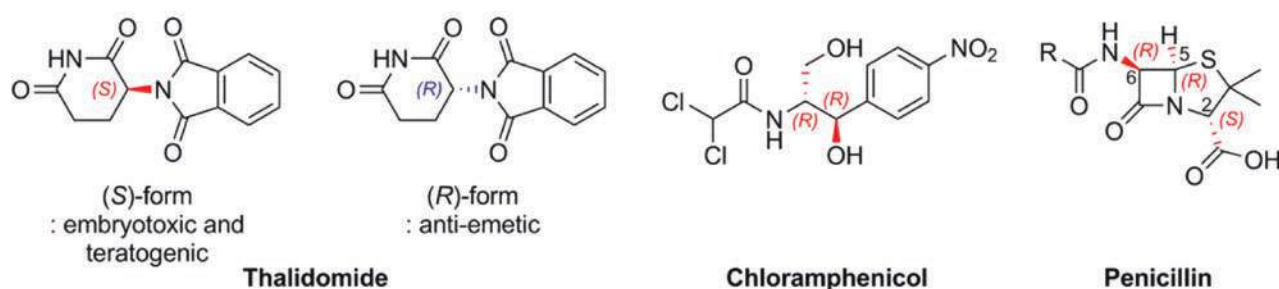
tion-based chemistry calculations” as an already error-prone process that must therefore be corrected, the “reduced coherence time requirements of the VQE make it more suitable for noisy quantum hardware, and they may enable the extraction of accurate results without the use of quantum error correction” [3] as one achievable solution. Verified experimentally “VQE can be inherently robust to some coherent errors, such as qubit over-rotation” but still suffer from noise-based accuracy [3], additional techniques have been proposed to solve the problem without recourse to such costly work-arounds.

Designed to mitigate errors rather than correct errors as a practical way for accuracy without extensive work and operational cost, akin to small but manageable computer bugs, the techniques as used through low-depth circuits to also hold the error rate correspondingly low, and these errors must consider mixed states rather than just pure states, as models assume but cannot provide perfected molecules, or perfect reactions.

Organics:

A Spring Board for Progress

As the subdiscipline of chemistry that studies spatial arrangement of atoms in molecules, stereoisomers with differing stereochemistry usually exhibit non-similar biological action in living things. Exemplified terribly by the thalidomide tragedy, where a treatment for morning sickness during pregnancy in the 1950s and early 1960s became deadly as it later was found that the

**Figure 1.**

Chemical structure of thalidomide, chloramphenicol, and penicillin with stereochemistry

mixture of two enantiomers was only partially safe; the S-form of thalidomide caused severe birth defects such as embryotoxicity, unlike the R-form (Fig. 1).

As critical to any biologically active molecule used in a process, including those outside an actual living system, Nuclear Magnetic Resonance (NMR) remains one way "...to assign the relative and absolute configuration of chiral molecules as well as their planar structures" for lab use, with molecular 3D structures as the essence of complete configurational analyses, made difficult by the flexible moieties of molecules over time [3]. Visible from the "...significant number of misassigned natural product structures" that already exist, a few methods exist to deal with the problem. X-ray crystallography "accomplishes full structural establishment of a molecule unambiguously" [3] but takes time and needs rather pure material that might not emerge in the course of tested combinations as littered with molecular byproducts. There is also the option of total synthesis of all plausible stereoisomers coupled with NMR spectroscopy, but even

AI as automating the process is a use of huge computing power and b of time and "becomes impractical especially when there are multiple stereogenic centers in molecules" [3].

Structural Maps, Computational Approaches

Consisting of 5 stages for small organic molecules (Fig. 2) that are "(1) obtaining pure compound by standard isolation techniques from natural sources or organic synthesis, (2) acquiring spectroscopic/ spectrometric data available (e.g. NMR, MS, UV, and IR), (3) planar structure characterization by 1D and 2D NMR data analysis (e.g. ^1H , ^{13}C , COSY, HSQC, and HMBC), (4) relative configuration assignment by additional NMR data analysis (e.g. coupling constants, JBCA and NOE/ROE correlations), and (5) absolute configuration determination using chiroptical methods (e.g. ECD/VCD, specific rotation) and/or chiral derivatization (e.g. Mosher's and Marfey's methods)", calculation of NMR properties are "effective for relative configurational establishment since these NMR properties tangibly differ depending on spatial

Structure elucidation of small organic molecules by contemporary computational chemistry...

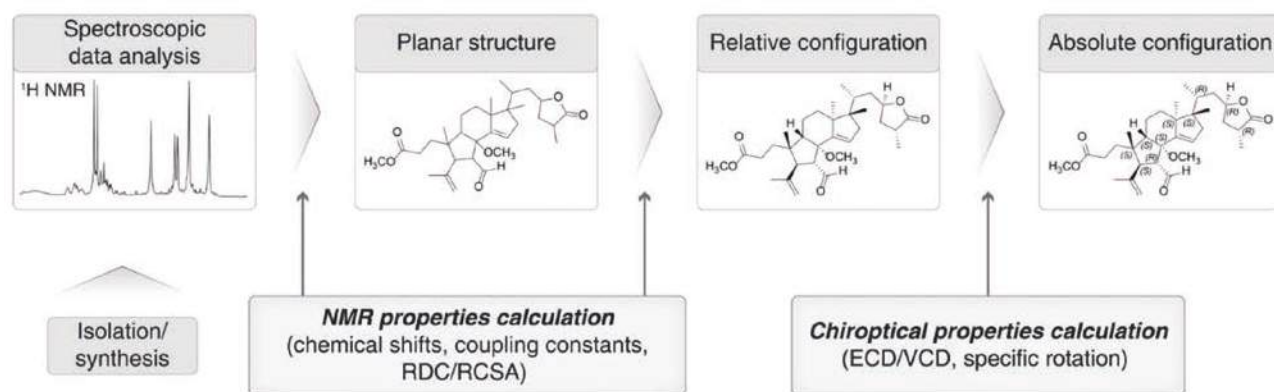


Figure 2.
General structural elucidation process for small organic molecules aided by computational methods

orientation of comprising atoms” [3] as added to/already found in useful areas of application.

Comparison of Calculated versus Experimental Values

Scaling Calculated NMR chemical shifts

Obtained previously, the unscaled values are first scrubbed with an additional “empirical scaling” process that removes systemic errors from NMR values prediction, which are then found by “plotting unscaled values of a set of training molecules against their experimental values” [3]. Seen by Tantillo et al. when they created the CHESHIRE repository that determines linear parameters for “commonly used NMR solvents, geometry optimization, and GIAO NMR calculation methods”, this very much facilitated NMR calculation involved in silico-related areas

of interest, with the process as done by the Goodman group, via their in-house databases for chemical shift calculations, seen below (Fig. 3) alongside the several applicable statistical parameters used in the process.

Solar Cells as the Future of Energy: Chemical Efficiency Required

Done with the theoretical and practical possibilities of how we might make use of organic molecules as low-energy method of delivering cargo with great efficiency possible, we now start the explanation of how such usability is applicable to the world as a future solar cell of great efficiency that can, if correctly done, perhaps solve our dependence on fossil fuels one day. With Organic Solar Cells (OSCs) “...considered to be one of the most promising photovoltaic technologies for the achievement of low-cost solar energy conversion because of their

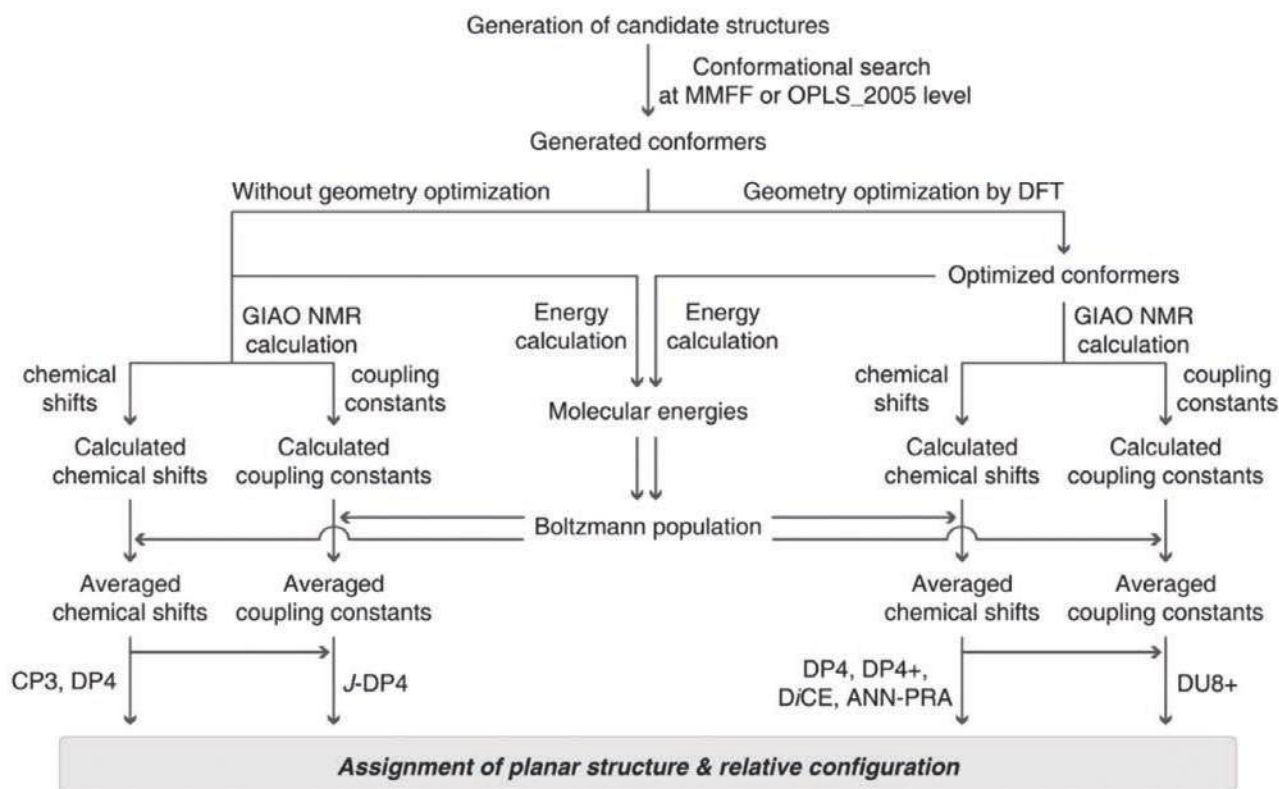


Figure 3.

Workflow of contemporary computational methods to assign planar structure and relative configuration of molecules

attractive properties of light weight, mechanical flexibility and the capability of large-area fabrication by roll-to-roll print method" [2], current ideas have three types available: Single-layer Schottky junction, bilayer and bulk heterojunction (BHJ) (Fig. 1).

With the basic structure being visible the same, the differences come from the difference in the organic layer, and how this layer reacts with others in terms of energy transfer etc. Made up of 4 main sections, from up being the glass substrate, the anode electrode mate-

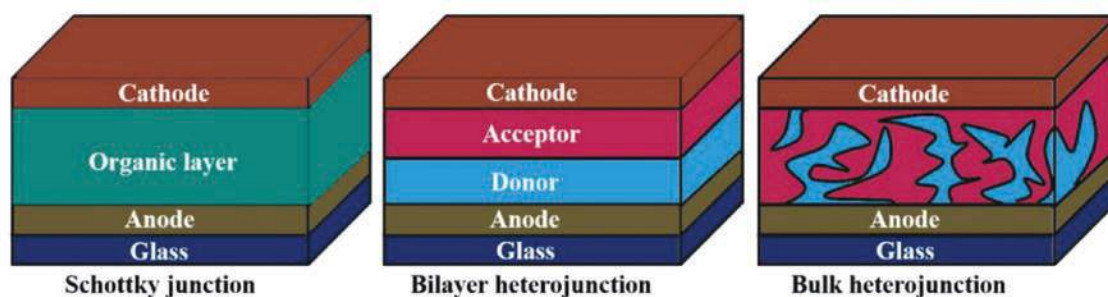


Figure 1.

Typical device structure diagrams of organic solar cells: Schottky junction, bilayer, and bulk heterojunction device.

rial, the organic active layer material, and the cathode electrode material, the organic active layer material is “placed between the cathode and anode to form a device with a special “sandwich” structure” that, for the BHJ, means that “the interface contact area to the donor (D)/acceptor (A) materials in the active layer is greatly increased” [3]. At the same time, the distinct characteristic of a good cross-network structure (that) can be formed in the blend films”, as... “conductive to the carrier transport in the active layer films” means that BHJs are the popular choice for current OSC ideas and development [2].

Despite work on the development of efficient polymer donors and fused ring electron acceptors for active lay-

er materials, the power conversion efficiency (PCE) of the state-of-the-art single junction OSC only reaches 18% [2]. As a good deal less than the most efficient artificial version at some 22-23%, the work that must be done will rely on the materials themselves that form the cell, rather than any underlying deficiency in how the cell is set up. Doable at a small scale that will allow nature imitation if molecular in size, organic photovoltaic materials, the research is now much aided by the confidence of automated AI, rather than the long times of trial and error previously needed (Figure 2) (although of course we do entirely dismiss the process as helpful for breakthroughs and further idea development).

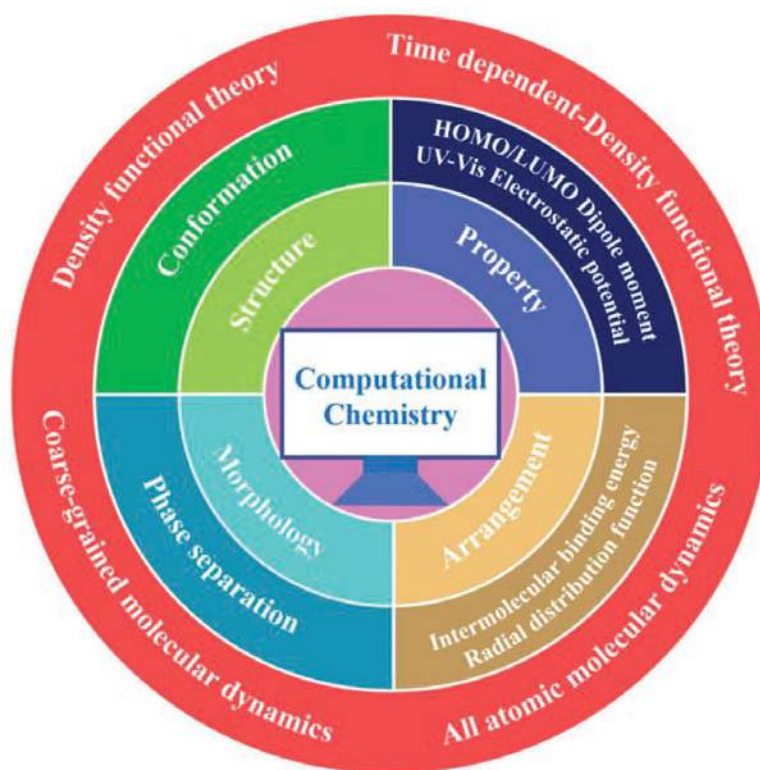


Figure 2.
Schematic diagram of typical theoretical methods and the applications of computational chemistry in organic solar cells.

Solving Chemical Deficiencies

Frontier molecular orbitals and UV

With the active layers of OSCs needing an acceptor with suitable levels of HOMO/LUMO to match the donor molecule for sufficient exciton dissociation and charge separation at the interface [2], HOMO/LUMO energy levels and the corresponding gaps of missing energy are closely tied to the electronic and optical properties that in turn affect OSC performance. Measured via cyclic voltammetry (CV), the energy gap found predicts the maximum absorption edge of the material, most efficiently done via computational simulation. With accuracy as dependent on the selected calculation method, the “difference between the fundamental gap and the optical band gap is defined as the exciton binding energy (EB)” [2] that causes the energy loss of the system, Wei and coworkers calculated a series of EB based on ITIC derivatives, finding that “the formation of three-dimensional stacking structure of small molecular acceptors was very important for reducing EB” [2].

With OSCs shown to rely on photon absorption as a main need of device efficiency, the best result is achieved when “absorption spectrum of organic photovoltaic materials should match with the solar radiation spectrum as much as possible” by being able to first theoretically, and later experimentally, predict and understand “the photoexcitation characteristics of organic photovoltaic materials” [2] as the new potential for perfected OSCs. From optical

parameters such as maximum absorption wavelength, to molecular orbital (MO) transition as contribution, it is the UV-Vis absorption spectrum range of the OSC active layers that “largely determines the output value of the photocurrent of the materials” [2] with emphasis on the relationship of molecular structure versus absorption spectrum.

With quantum computing now the standard to estimate the UV-Vis absorption spectrum of active layer materials via electronic transitions and the contribution of orbital transitions, results are now much faster than older methods, Fréchet et al. having designed and synthesized symmetrical dinuclear platinum (II) oligomers (BTD-Pt-T(n)) based on thienyl-benzothiadiazole-thienyl (BTD), using them to create a “unique absorption peak of the D-A structure with a narrow band gap”, as needed for efficient OSCs; the literature reports that “energy level hybridization of the donor and acceptor units result in a particularly low HOMO-LUMO separation in the D-A systems” [2] that thus leads to increased energy efficiency through closely linked levels of minimum energy levels.

As also possible for acceptor materials, the UV-Vis absorption simulation for OSCs is not just dependent on donor materials; both sides must be balanced out correctly for a more ideal result. Predicted through calculation, structural properties of acceptor molecules save cost and speeds experiments, one example seen from Hou et al., on their chlorinated low bandgap acceptor BTP-4Cl that replaced halogen atoms

of the fluorinated non-fullerene acceptor BTP-4F. With a match when the “theoretical simulation of the BTP-4X absorption spectra revealed that the maximum absorption peak of BTP-4Cl was red-shifted than that of BTP-4F, which was consistent with the experimental test and proved the rationality of the molecular design”, the potential of such molecules as useful for OSC, as BFTT-N and BFTT-TN “have higher absorption intensity attributing to the larger sum of during transition processes, and the maximum absorption peak of BFTT-TN have a red shift compared with ITIC, which indicates that the strong absorption of quad-rotor-shaped fused-ring electron acceptors (FREAs) can improve the device performance” [2], is very promising for organics as possible replacements as computing becomes more accurate, with the design of high-performance acceptor molecules being a much more likely outcome for organics than donor molecules, as the taking of energy is for OSCs much more significant to usability than where the source of energy is transferred from (considering that plants absorb a tiny percentage of sun, but the sun itself provides limitless but unharnessed energy potential).

Electrostatic potential

The second point of importance for design of a better OSC, the play of nuclei and electrons of a molecule as existing electrostatic potential that “can be used to analyze charge generation, intermolecular electrostatic interaction, reaction sites and molecular proper-

ties” as part of intrinsic energy density and transfer possibility based on electron movement, is very promising. With $V(r)$ in any particular region of space depending on dominance of either nuclei or electrons, electronic density as determined by a diffraction test and theoretical calculation becomes an early stage OSC consideration, with fullerene and derivatives as acceptors via their energy offset between the donor and acceptor materials as ensuring sufficient levels of charge generation in the system [2].

As the charge transfer state energy (ECT) and the bandgap energy (E_g) of the OSC devices results in a large voltage loss, one solution has been the generation of “charges in a material system with negligible driving force especially in the non-fullerene-based OSCs” [2] as a solution, with molecular ESP as revealing charge generation mechanisms in OSCs by Hou et al. Known that “ π -electrons and holes in organic photovoltaic materials are delocalized in their π -orbitals, and the exciton dissociation in the blend system occurs through the orbital overlap at the interface of donor and acceptor” [2], the ESP distributions of the simplified polymer donor 2 and the non-fullerene acceptor IT-4F' were calculated to figure out the average ESP of each atom, followed by the result that “the intermolecular electric field (IEF) formed at the interface between IT-4F' and (DCBT-2F)2 may facilitate exciton dissociation and promote the transfer of electrons from the donor to the acceptor” [2] as a base for improved OSCs that use natural molecules with

almost maximum potential for energy discharge and acceptance, as able to achieve much higher rates of efficiency than the mere 18% that now holds true.

To completely verify the efficiency of the energetic offset between fullerene and non-fullerene systems as reliant on the average ESP of fullerene C60, simulation for it showed that “the ESP of C60 is distributed over the entire spherical surface, and the average ESP value is lower than that of IT-4F”, which allows for the IEF at the interface between the donor and fullerene, as weaker than that between the donor and IT-4F’, as perhaps the reason for the non-fullerene system as highly energy efficient, any energy offset being negligibly minor [2].

In order to verify “that non-fullerene OSCs generate charge with negligible energy offset as driving force”, Hou et al. used PTO2, IT-4F and PC71BM as donors and acceptors, respectively, to test calculations from the perspective of chemical structure (Fig. 6b). The surface of “PC71BM shows positive ESP, but its value is much lower than that of IT-4F” [2], resulting in the molecular interface of PTO2:IT-4F as significantly greater than that found between PTO2:PC71BM. To demonstrate the effect of IEF on exciton separation, charge density distributions of the lowest excited state of the two systems were simulated and analyzed with a fixed face-to-face distance (Fig. 6b), seen that “the electrons and holes are mainly distributed on the PC71BM and

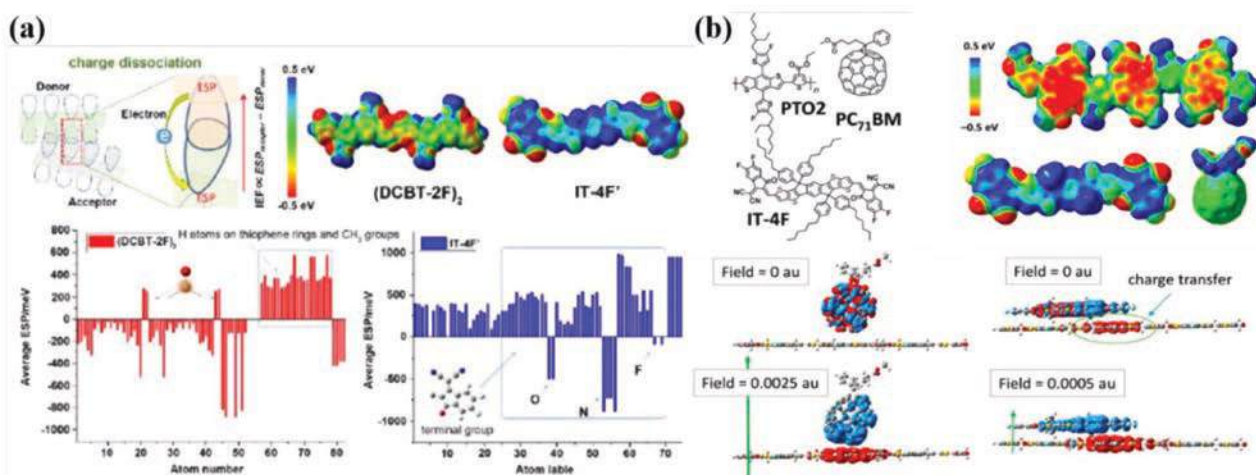


Figure 6.

(a) Exciton dissociation process occurs at the donor/acceptor interface, and the ESP difference can form an intermolecular IEF. ESP distributions on the polymer donor (DCBT-2F)₂ model and the non-fullerene acceptor model IT-4F. Average ESP statistics of each atom on the polymer and acceptor models. Reproduced with permission from ref. 105. Copyright 2018, Wiley-VCH.

(b) Chemical structures of the polymer donor, fullerene, and nonfullerene acceptors. ESP distributions of the donor and acceptors. The electron/hole density distributions of the lowest excited states without/with the additional electric field between PTO2 and PC₇₁BM, as well as PTO2 and IT-4F; red and blue represent hole and electron densities, respectively. Reproduced with permission from ref. 106. Copyright 2019, American Chemical Society.

highly overlapping in the PTO2:PC71BM composite”, but are instead distributed on the donor PTO2 in the PTO2:IT-4F system, meaning that the acceptor has undergone charge transfer after excitation [2].

With “the increased charge density delocalization may be related to a larger average ESP on the surface of acceptor”, PBDB-TF:IO-4Cl as a way forward for non-fullerene OSCs help via the larger ESP difference versus that of the PBDB-TF:ITIC system, the ESP difference of non-fullerene-based OSCs furthered even more when Hou et al. Made use of D-A conjugated polymer donors with different conjugated length of the D-units in their backbones; the backbones of these “three polymers show similar planar zigzags, and the proportion of contributing negative ESP parts increases with the extension of the D unit, which improves the donor/acceptor orbital overlap to facilitate charge transport” [2], and with new materials and device processes for newer OSCs with better stability, insensitive film thickness and complementary light absorption ranges, candidates for much better energy absorption are now quite a bit more broad than just a few years ago. As opportunity expands, such as the efforts of Professor Eugenia Kumacheva of the University of Toronto and partners as now having demonstrated attached chiral molecules to the surface of nanoparticles that are organized to create hybrid building blocks in ensembles with excellent chiroptical properties that are perfect for organic molecule usage by

being controllable and also predictably self-assembling [5], little remains to be said for possible research, save that it continues to make very good progress.

Conclusion

Energy efficiency. As the main concern of humans nowadays, one way to solve the problem is the use of chemical synthesis, by use of minimum-energy structures that are both organic, and tied to the idea of sustainable but also scalable production. Achievable by the available quantum computing as able to speed up the process of testing out minimum energy levels found in molecules as synthesizable compounds that can be made organic to very high efficiency via OSCs as the new wave of technology that can solve the dependence on fossil fuels, the entire process has become possible due to modern AI as the automatic process that can test a huge number of variables without confusion while also keeping a detailed record of which results yield the best solutions to the problem at hand. With chemistry as powered by naturally-guided molecules and artificial computing as the needed speed input, methods of energy production are currently undergoing an overdue transformation that is not impossible, but does require involvement from both scientists, computer and chemist, to do so.

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**COMPUTATIONAL CHEMISTRY: MODELING TO OBTAIN MINIMUM
ENERGY STRUCTURES AND REACTION PATHWAYS**

Alp Baran ERKUL*

Abstract

Although chemistry investigates the most basic aspects of life, such as particle properties, it can improve much more with the use of technology. Computational chemistry enables modeling to obtain minimum energy structures and reaction pathways. This result is very important in different aspects of life. Especially in industry, where fast production is important, the aim is to develop systems to obtain fast reaction pathways. In this research paper, the wide applications of artificial intelligence in chemistry are exemplified. The university researches are investigated to see whether technology can be integrated to get beneficial outcomes. Moreover, the comment of many professors and researchers about integrating AI into chemistry are investigated. Both the advantages and challenges are also considered, but it is noticed that the advantages will outweigh the disadvantages. Especially, in the 21st century when the primary aim is the fast production, obtaining minimum energy structures will be of highest importance. Therefore, it is argued that integrating artificial intelligence in chemistry should be priority for getting beneficial and fast results.



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